Title

Hotel Management System

Abstract

This research paper presents the design and development of a Hotel Management System for Anant Luxe Resort. The system aims to automate hotel operations such as room booking, food ordering, billing, and customer data management. Traditional hotel systems rely on manual processes, which are time-consuming and prone to errors. This project introduces a digital solution using web technologies like HTML, CSS, and JavaScript. The system also integrates QR code-based payment for faster transactions. The results show improved efficiency, reduced workload, and enhanced customer satisfaction. The study highlights the importance of digital transformation in the hospitality industry.

Introduction

The hospitality industry is rapidly evolving with the use of digital technologies. Hotels require efficient systems to manage reservations, customer records, and billing processes. Manual systems are not only slow but also increase the chances of human error.

The Hotel Management System developed for Anant Luxe Resort provides a modern solution to these problems. It allows customers to book rooms online, order food, and make payments easily. At the same time, it helps hotel staff manage operations efficiently.

This project focuses on creating a user-friendly, reliable, and efficient system that enhances both customer experience and hotel productivity.

Literature Review

Various studies and research papers highlight the importance of hotel management systems in improving operational efficiency. Earlier systems were mainly desktop-based and limited in functionality. However, modern systems are web-based and provide real-time services.

Key observations from previous studies:

- Automation reduces human errors and saves time
- Digital systems improve customer satisfaction
- Online booking systems increase hotel revenue
- Integration of payment gateways enhances convenience

Researchers have also emphasized the need for secure and scalable systems in the hospitality sector.

Methodology

The development of the Hotel Management System followed a structured approach:

🔹 **1. Requirement Analysis**

- Understanding hotel needs
- Identifying features like booking, billing, and admin control

🔹 **2. System Design**

- Designing user interface
- Planning database structure
- Creating system flow

🔹 **3. Development Tools**

- HTML for structure
- CSS for styling
- JavaScript for functionality

🔹 **4. Implementation**

- Creating booking system
- Adding food ordering module
- Integrating QR payment system

🔹 **5. Testing**

- Checking all buttons and features
- Fixing errors and bugs

Results

After successful implementation, the system showed the following results:

- Faster room booking process
- Accurate billing system
- Easy food ordering interface
- Smooth payment experience using QR codes
- Efficient management of customer data

The system worked effectively in reducing manual workload and improving overall efficiency.

Findings

The research identified several important findings:

- Digital systems significantly improve operational speed
- Customers prefer online booking and payment options
- Automation reduces dependency on manual work
- User-friendly interface increases system usability

Discussion / Analysis

The developed system proves that technology plays a crucial role in modern hotel management. Compared to traditional methods, the digital system provides better accuracy and efficiency.

However, some challenges were observed:

- Need for internet connectivity
- Basic level security implementation
- Limited advanced features

Despite these limitations, the system is highly effective for small to medium-sized hotels.

Conclusion

The Hotel Management System for Anant Luxe Resort successfully achieves its objective of automating hotel operations. It improves efficiency, reduces errors, and enhances customer satisfaction. The system demonstrates how digital solutions can transform traditional hotel management practices.

Future improvements can include mobile applications, AI-based recommendations, and advanced security features.

References / Bibliography

- Books on Hotel Management Systems
- Online tutorials for HTML, CSS, JavaScript
- Research articles on hospitality management
- Educational websites like W3Schools